

Where to Stop: Measuring the Value of Spatial Early Exit in Learned Image Decoding

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Abstract

001 *Spatially adaptive decoding can save computation by stop-*
002 *ping easy regions early, but a large total saving need not*
003 *imply a large benefit from content adaptation. We study*
004 *this distinction in the intra decoder accompanying DCVC-*
005 *UF. A shared synthesis trunk feeds tile-level exits, pointwise*
006 *adapters and a full-frame reconstruction head, while the*
007 *analysis transform and entropy model remain fixed. We com-*
008 *pare source-informed exit search and a learned router with*
009 *uniform depth and ordered dithering on the same recorded*
010 *mixed reconstructions. Across 53 first intra frames and five*
011 *quality indices, the router saves 27.97% of modelled de-*
012 *coder multiply-accumulates at a 0.1 dB target, compared*
013 *with 25.43% for dithering on 263 common frame-quality*
014 *pairs. The paired advantage is 2.54 percentage points and*
015 *falls to 0.38 points at a 0.3 dB target as the exit ladder sat-*
016 *urates. A reference audit shows that selection and final*
017 *reporting use different reconstruction anchors, preventing a*
018 *delivered-quality guarantee from the nominal target alone.*
019 *Historical timing isolates a further cost: acquiring source-*
020 *error tables at the encoder. These findings identify where*
021 *spatial routing adds value and which quality, signalling and*
022 *execution costs must be included before arithmetic savings*
023 *can be interpreted as practical acceleration.*

024 1. Introduction

025 A learned decoder applies the same synthesis network to
026 every region of an image. A fixed network makes execution
027 predictable, but it also spends the same depth on regions
028 with different reconstruction requirements. Spatial early exit
029 offers a direct alternative: retain a shared representation and
030 allow individual regions to stop once further computation
031 has little value. The central question is how much this con-
032 tent dependence contributes beyond simply using a cheaper
033 decoder.

034 This distinction matters because several mechanisms can
035 produce a similar headline saving. A uniform shallow net-

work removes work everywhere. A content-blind mixture of
two depths makes the available cost levels finer. A learned
router additionally decides *where* the more expensive depth
is useful. Comparing only with a full-depth reference cred-
its the router for all three effects. It does not establish that
content prediction is worth its own execution and calibration
costs.

We investigate this question in the intra decoder sup-
plied with DCVC-UF [5]. Our shared-latent implementation,
FLEX, divides the synthesis trunk into block pairs and at-
taches tile-level exits (Fig. 1). A pointwise adapter aligns
each exiting feature with a shared reconstruction head. A
full-frame repair operation addresses the known tile grid.
The encoder and entropy model remain fixed, but the syn-
thesis network is fine-tuned: compatibility of the coded
representation does not require the reconstruction network
to remain unchanged.

The compression setting provides two distinct sources of
decision information. An encoder can compare candidate
reconstructions with the source image. A decoder can use re-
covered latents, entropy-model information and intermediate
features, but does not know the source error. These regimes
must remain separate. In particular, a router with decoder-
available inputs is not an autonomous decoder-side policy if
its control parameter is fitted to the source of each test image.
Our recorded budget sweeps use that latter, source-calibrated
protocol, and we label them accordingly.

Three observations organise the study. First, routing adds
a measurable but budget-dependent advantage over ordered
dithering. At a 0.1 dB target the paired gain is 2.54 percent-
age points; at 0.3 dB it is 0.38 points. Second, selection and
reporting use different reconstruction anchors and spatial
supports, so a nominal table target cannot be read as a deliv-
ered guarantee. Third, source-informed decisions have an
encoder cost: the required error table contains reconstruc-
tions, not merely cheap router logits.

Our contributions are a shared-latent spatial early-exit
implementation; a paired evaluation that separates depth
reduction, blind mixing and learned allocation; and a mea-
surement analysis connecting the quality target to delivered

distortion and the cost of obtaining routing information. The current evidence supports decoder arithmetic reductions and historical stage timings. It does not yet establish an end-to-end speed advantage over the best content-blind policy. Independently trained depth-reduced codecs form a separate, ongoing experiment rather than additional exits of this model.

2. Related Work

Spatially adaptive reconstruction. ClassSR routes super-resolution patches to networks of different capacity [4]; APE instead permits patches to exit a shared network at different depths [10]. RBQE also uses early exit to adapt the effort spent enhancing already compressed images [11]. Building on spatial capacity selection, we examine early exit inside a learned synthesis transform and the constraints of its coded representation: which decision information is available at each endpoint, what must be signalled, and how a local assignment affects final reconstruction quality.

Adaptive complexity in compression. Slimmable compressive autoencoders provide multiple model widths [12]. AdaNIC combines blockwise transform capacity with a routing agent [8], while cgSlimDecoder selects decoder channel widths under complexity constraints [2]. These methods already establish content- or complexity-dependent computation within learned compression. Our comparison focuses on depth allocation in the synthesis path of a fixed coded representation and on its incremental benefit over blind mixtures. A matched reproduction of these alternatives is the next step in evaluating the broader adaptive-codec frontier.

Spatial competition between complete codecs. Blard et al. [1] select a specialized codec per region using source rate–distortion cost and transmit a mode map. This is a close precedent for our ongoing independent codec-bank direction, where each expert has its own encoder and entropy model. The present shared-latent experiment instead varies synthesis depth after one common representation. The two settings require different baselines and different signalling claims.

Efficient synthesis and practical execution. Shallow decoders investigate how computation can be shifted toward the encoder [13]. DCVC-RT distinguishes arithmetic cost from operational overhead, including memory traffic and function calls [3]; DCVC-UF further develops a throughput-oriented video codec [5]. Recent practical image-codec design also optimises perceptual performance against measured device runtime [9]. These results motivate reporting actual execution alongside multiply–accumulates. Our scope is the supplied intra model, not DCVC-UF’s inter-frame or chunk-coding performance.

Distortion and perception. Generative compression explicitly studies reconstruction fidelity and perceptual realism [6, 7]. A small PSNR change therefore cannot establish perceptual equivalence. We use archived non-DCVC experiments only to examine this limitation and the dependence on decoder architecture; they are not evidence of a perceptual guarantee for FLEX.

3. Spatial Early Exit under a Quality Target

3.1. Shared representation and conditional execution

Let $\hat{y}$ denote the decoded latent of a fixed analysis transform and entropy model. A full-frame upsampling stage and four shared trunk blocks produce features with 384 channels at one eighth of the RGB resolution. The pinned e15 configuration partitions this canvas into 32×32 feature tiles, corresponding to 256×256 RGB regions. The remaining eight blocks are executed in pairs. Although the model exposes six nominal exit indices, the first two lie before the split and are clamped: inference has four distinct reachable depths, $\mathcal{K} = \{6, 8, 10, 12\}$.

Each active tile traverses the next pair of blocks. Tiles assigned to that depth pass through an exit-specific pointwise adapter and leave the active batch; the others continue. The adapted tiles are stitched into a feature canvas. A grid-conditioned repair and the shared output head reconstruct the image. The halo-free tiled trunk uses replicate padding. This reduces work only when the execution path actually removes departed tiles; evaluating all branches and masking their outputs would have different runtime behaviour.

3.2. Selection objective and its approximation

For tile t and exit k , let $M_{t,k}$ be the source MSE measured from a candidate tiled reconstruction, and let R be the source MSE of the full-depth released-weight reference on the padded source. The search uses the table surrogate

$$\tilde{\Delta}(\mathbf{k}) = 10 \log_{10} \frac{N^{-1} \sum_{t=1}^N M_{t,k_t}}{R}. \quad (1)$$

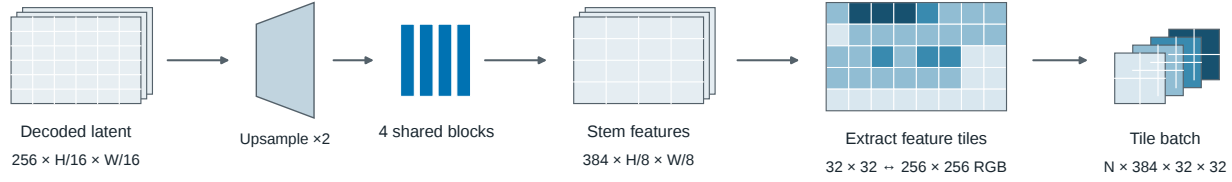
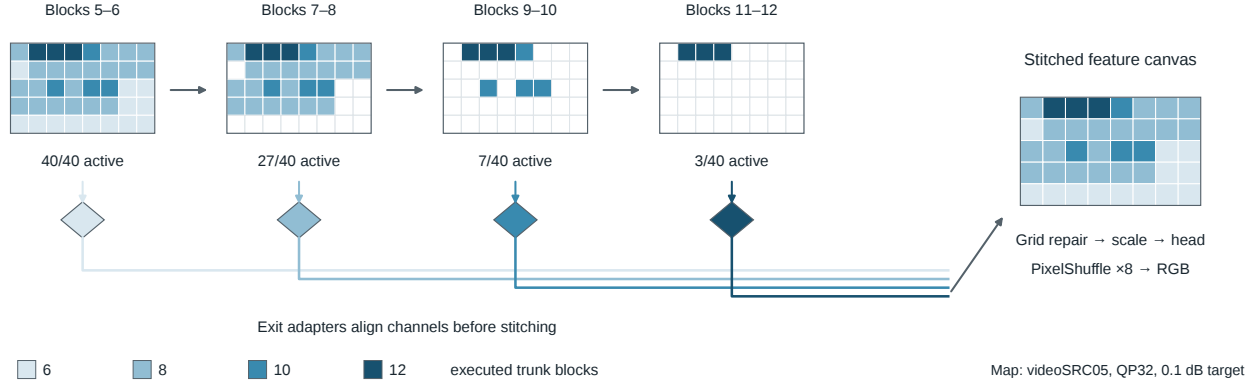
The recorded cost c_k is normalised to a full-frame reference decode. For equal padded tiles, the cost model gives

$$C(\mathbf{k}) = N^{-1} \sum_t c_{k_t}, \quad S(\mathbf{k}) = 100 [1 - C(\mathbf{k})]. \quad (2)$$

Source-informed selection chooses

$$k_t(\lambda) = \arg \min_k \{M_{t,k} + \lambda c_k\}, \quad (3)$$

with a per-frame search for the quality target. We call this procedure *source-informed search*, rather than a certified global optimum of the final reconstruction problem.

a A shared representation, spatially variable synthesis**b Only surviving tiles traverse the next block pair****Frozen encoder + entropy model**

The synthesis network and exit adapters are fine-tuned.

Shared head couples neighbouring tiles

Evaluate the final mixed reconstruction after cropping.

Figure 1. Shared-latent spatial early exit. Four trunk blocks run over the full feature canvas. The four reachable exits execute 6, 8, 10 or 12 total trunk blocks. The recorded router assignment controls which tiles continue through each block pair; blank cells have already exited. Pointwise adapters feed a stitched canvas, followed by grid repair and a shared head. Tensor values are schematic; tile assignments are recorded data from videoSRC05 at QP32 and a 0.1 dB target. This is distinct from the ongoing independent D2/D4/D6 codec training.

The distinction is substantive. Tables are selected on the padded RGB image; reported loss uses the final mixed reconstruction and the full-frame *fine-tuned e15* reconstruction, both cropped to the source extent. The two anchors differ in synthesis weights as well as spatial support; the supplement traces both evaluation paths and the checkpoint comparison. In addition, full-frame repair and head operations couple neighbouring tiles. Consequently, $\tilde{\Delta}$ is a selection surrogate, not an identity for delivered distortion. We report both the target and the measured loss and count crossings of the nominal threshold explicitly. These crossings do not isolate enforcement error under a common reference; that requires a new evaluation against both cropped anchors.

3.3. Router and content-blind controls

The router projects intermediate synthesis features and latent/entropy-scale information, pools them per tile and predicts exit scores. Given log scores $\ell_{t,k}$, the recorded

decision is

$$k_t(\beta) = \arg \max_k \{\ell_{t,k} - \beta c_k\}. \quad (4)$$

The evaluated sweep fits β separately on each source frame's table. This measures the attainable allocation with source calibration. Deployment with a fixed, held-out control parameter is a separate experiment. Increasing β makes the selected cost non-increasing, but need not make source distortion monotone. The supplement audits the recorded bisection against the finite set of assignments along this control path.

Uniform depth selects one reachable exit for the whole frame. Bayer dithering mixes two adjacent exits according to a fixed spatial ordering, with its mixture fraction selected under the same table-based quality target. It has no content predictor but can realise finer average costs than uniform depth. All four methods use the same checkpoint and per-exit cost vector.

3.4. Rate and decision information

Keeping the coded latent fixed does not make all routing modes byte-identical. Encoder-selected maps require a communicated decision representation. A source-calibrated decoder router also needs its control parameter to be communicated or otherwise reproduced. The curves in this study model decoder MACs and do not add router cost or decision signalling to that quantity. They therefore isolate allocation quality; a deployment comparison must add those costs and verify actual encoded bytes separately.

4. Results and Measurement Analysis

4.1. Protocol

The primary archive contains the first intra frame of 53 CTC sequences at QP values $\{0, 16, 32, 48, 63\}$, giving 265 frame-QP pairs. We evaluate the pinned e15 shared-latent checkpoint and one fixed router checkpoint at six target budgets between 0.05 and 0.5 dB. Reconstructions were produced by the mixed tiled decoder; this revision reanalyses those outputs and does not introduce new GPU measurements. Means compare the common cases for which all four rules returned a plan. This leaves 204 pairs at 0.05 dB, 263 at 0.1 dB and 265 at each larger target. Missing plans are not silently treated as successful full-depth fallbacks.

Paired confidence intervals resample sequences with all their available QPs kept together, using 5,000 bootstrap draws. They describe variation over the available content at fixed trained weights, not training-seed uncertainty. We also repeat the budget trend on the same 204 pairs throughout to expose the effect of changing cohort membership.

4.2. The incremental value of content prediction

At a 0.1 dB target, source-informed search saves 29.52% of modelled decoder MACs, the router 27.97%, Bayer dithering 25.43%, and uniform depth 21.07% (Fig. 2). The router's paired advantage over dithering is 2.54 points, with a 95% interval of $[1.83, 3.26]$. This is the incremental arithmetic value of learned allocation under the recorded calibration procedure, not the router's total saving against full depth.

The advantage declines as the target relaxes: it is 1.31 points at 0.2 dB, 0.38 at 0.3 dB and 0.02 at 0.5 dB. The cheapest reachable exit imposes a 39.12% ceiling on this cost model. Near that ceiling, many maps assign almost every tile to the same exit, leaving little decision value for a content predictor. The fixed-cohort analysis retains the declining trend, and the QP breakdown shows that its location depends on the operating range.

A fixed-histogram diagnostic isolates spatial placement further. For each recorded map, we compute the exact expected table MSE after uniformly permuting its exit assignments. This preserves the number of tiles at every depth and hence its modelled cost. At 0.1 dB, the router's placement

improves the table statistic by 0.0213 dB on average (95% interval $[0.0167, 0.0262]$), while dithering gives 0.0007 dB $([-0.0002, 0.0017])$. These are source-table diagnostics, with the derivation and fixed-histogram optimum in the supplement; final mixed reconstructions would need to be decoded separately.

4.3. A quality target is not a delivered guarantee

The policies do not achieve exactly the same loss at the same nominal target (Fig. 3a). At 0.1 dB, the router exceeds target plus 10^{-4} dB in 7/263 cases in RGB and 49/263 cases in YUV 6:1:1. Maximum losses are 0.110 dB and 0.291 dB, respectively. These are crossings of the nominal numeric target in the exported metric, not a clean test of a common-reference constraint: selection uses a released anchor on padded RGB, while reporting uses the fine-tuned full-depth output on cropped images. Metric choice and cross-tile coupling are additional differences. The exported records cannot separate their contributions. A framewise guarantee requires the same anchor, support and metric in selection and final verification.

4.4. The encoder cost of knowing the source error

The error table in Eq. (3) requires candidate reconstructions. The historical QP32 benchmark recorded 1,347.65 ms for six trial exits, 1,023.26 ms after deduplicating the four reachable outputs, and 336.87 ms when a single tiled pass reused the common prefix (Fig. 4). The last change reduces table-construction time by approximately fourfold. These are the archived upper medians of 24 pooled trials over eight frames, not an end-to-end encoder speedup. The raw repetition distribution and device load history were not retained. Moreover, the archived GPU search timer can include pending reference-decode work, so we do not use it as an isolated search latency.

A fixed-control router can avoid the per-frame source-error table. The source-calibrated router curves reported here cannot: they use that table to choose β . A useful next comparison must therefore measure the complete decision path under each information regime, including any extra stem work at the encoder, map/control signalling and the final quality check.

4.5. Spatial assignments and transfer limits

Figure 5 shows the recorded maps of one QP32 frame at two targets. The example is chosen from the five archived thumbnails by proximity to the aggregate router-dither premium at 0.1 dB. The source thumbnail supplies spatial context; it is not presented as a reconstruction-quality comparison. The convergence of maps at the relaxed target illustrates the ladder's saturation, but the aggregate claim comes from the full paired analysis.

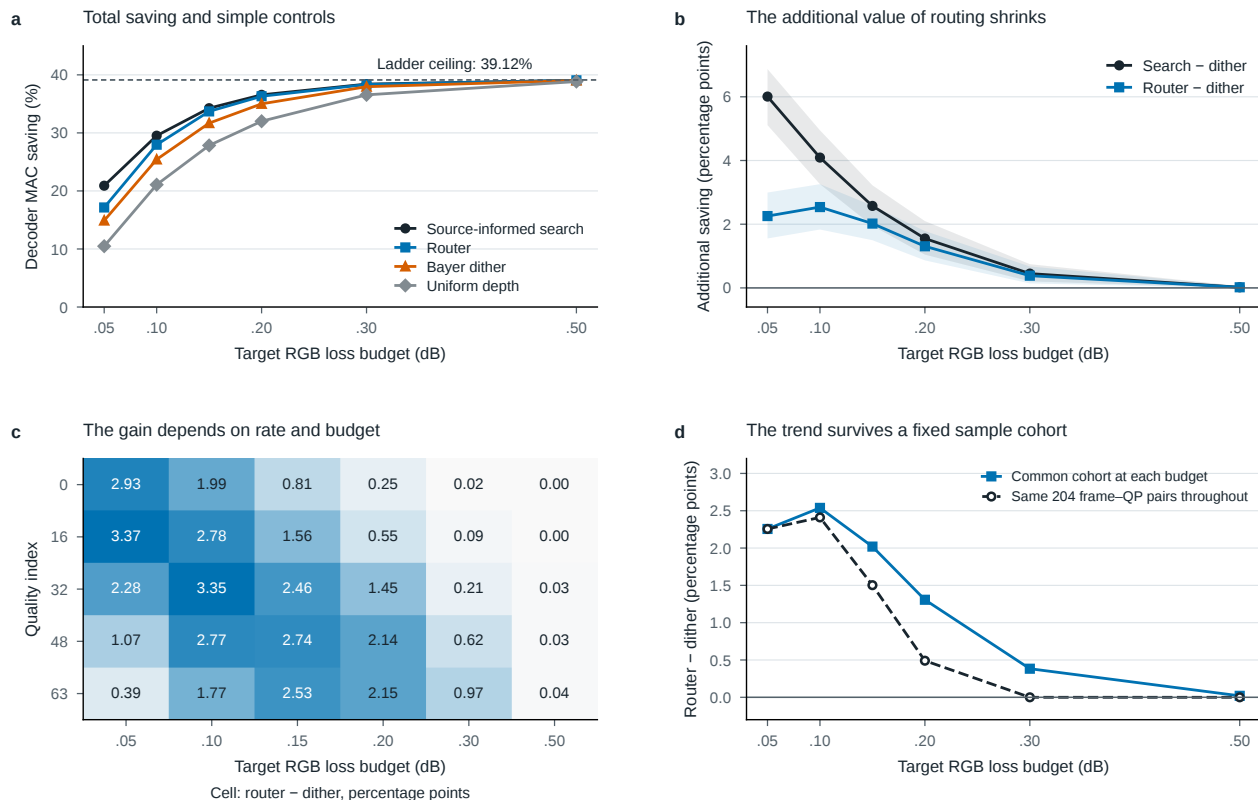


Figure 2. **Separate total saving from the value of routing.** **a**, Four policies on their common cases at each target. **b**, Paired differences with 95% sequence-cluster bootstrap intervals. **c**, Router-dither difference by QP and target, in percentage points. **d**, The same comparison on a fixed cohort of 204 frame-QP pairs. All curves use source-calibrated per-frame controls and a decoder MAC model; they do not include router/signalling cost or constitute latency measurements.

Archived experiments on other decoders provide a useful boundary check (Fig. 6). Their selection criterion measures deviation from a reference reconstruction, and their region/halo accounting models potential compute reductions. The dense masked implementation evaluates candidates before selection; these results therefore do not demonstrate conditional execution speedups. The source-informed heuristic does not uniformly beat the best blind control. Separate Kodak perceptual-budget experiments also show that a PSNR constraint can permit appreciable LPIPS degradation. Neither finding can be transferred into a numerical claim about DCVC-UF without a matched experiment.

5. Discussion and Next Experiments

The relevant deployment question is whether content prediction improves the rate-distortion-time frontier after its own costs are included. The current paired analysis identifies a region of budgets where further routing work is plausible and a saturation region where even an accurate predictor has little room to help. It also gives strong controls: uni-

form depth, ordered dithering, and source-informed selection should accompany every router comparison.

The margin also sets an overhead budget. Under the explicit simplifying assumption that synthesis time is proportional to the recorded MAC model, the 0.1 dB router-dither difference permits extra router work equivalent to only 3.40% of the dither synthesis time before the advantage disappears. At 0.3 dB that allowance falls to 0.62%. These are compute-based break-even illustrations, not measured latency predictions; real grouped execution and memory traffic determine the actual margin.

Three experimental gaps are decisive. First, quality must be verified on the final cropped reconstruction in the chosen metric, with a measured fallback for infeasible or over-budget plans. Second, a held-out, decoder-available control policy must be compared separately from test-frame source calibration. Third, actual encoded bytes and end-to-end timings must include decision signalling, entropy decoding, routing, tile movement and stitching, measured on the same device with paired execution orders.

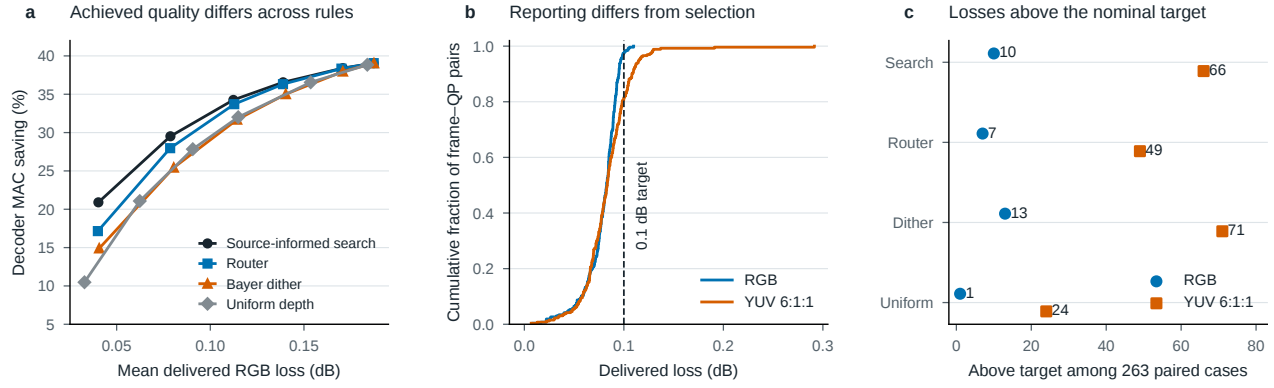
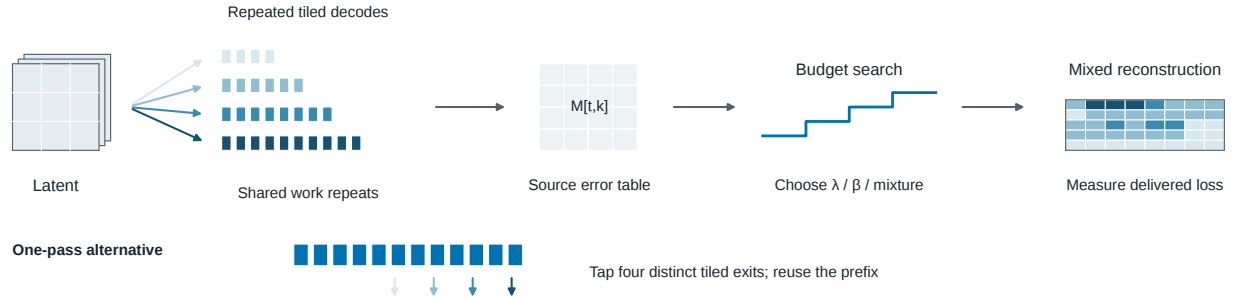
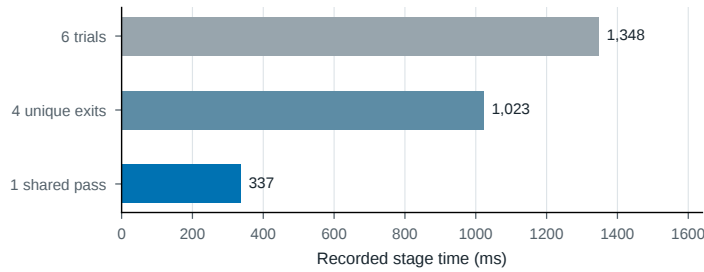


Figure 3. **Measure the reconstruction that is delivered.** **a**, Recorded mean RGB loss against MAC saving; connected segments guide the eye. **b**, Empirical loss distributions for the router at a 0.1 dB target. **c**, Losses above the nominal target among 263 paired cases, measured after cropping, with tolerance 10^{-4} dB. The search used padded RGB tables with a released-weight anchor; reporting uses the fine-tuned e15 full-depth anchor. Counts are not an isolated guarantee test. No verified fallback was applied in this archive.

a The source is available at the encoder, but the error table must be computed



b Recorded table construction time



Historical timing: 8 frames $\times$ 3 repetitions at QP32. No new end-to-end latency measurement.

c Calibration changes the cost

Fixed, held-out β

Router uses decoded features.
No per-frame source table is required.

Per-frame source-calibrated β

The plotted budget sweep uses $M[t,k]$.
A fast router does not remove that cost.

Figure 4. **Obtaining the decision information has a cost.** **a**, Candidate reconstruction, source-error tabulation, control search and final mixed reconstruction are distinct stages. A shared pass can reuse the trunk prefix. **b**, Archived table-construction timings; no raw trial uncertainty is available. **c**, Fixed held-out control and per-frame source calibration have different information requirements. These historical stage measurements do not establish current end-to-end latency.

a Spatial decisions converge as the quality budget relaxes

videoSRC05_1920x1080_25.yuv · QP32 · archived source luma

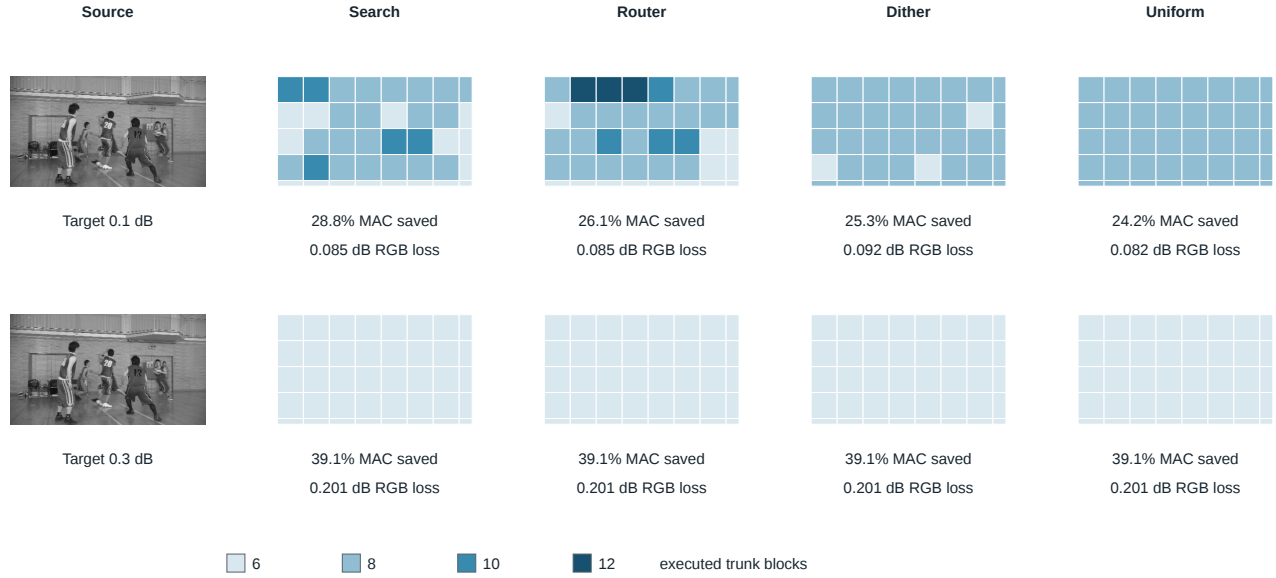


Figure 5. **Recorded spatial decisions under two budgets.** The left column is source luma. Other columns show actual maps, cropped to the visible image extent, with executed depth, modelled MAC saving and delivered RGB loss. This figure visualises allocation rather than perceptual equivalence. The example-selection rule is stated in the text.

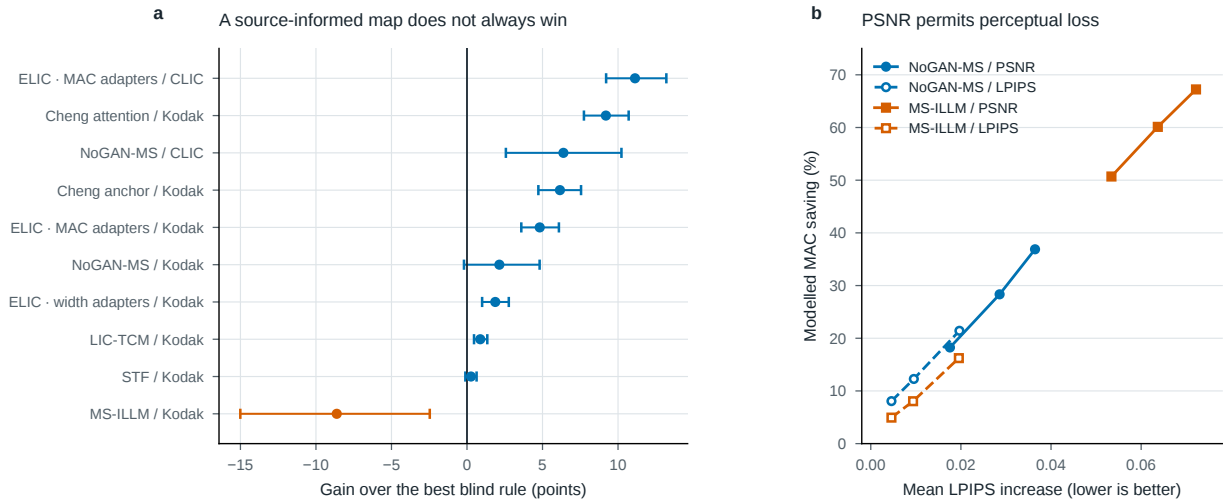


Figure 6. **Architecture and metric limit the scope of the result.** **a**, Source-informed selection minus the best dataset-level blind rule at a 0.1 dB target, with paired image-bootstrap 95% intervals. Adapter variants and datasets remain separate. Savings are modelled over dense masked reconstructions. **b**, Independent Kodak-24 evaluations under PSNR or LPIPS constraints for NoGAN-MS and MS-ILLM. The three points per curve are recorded operating points, not interpolated observations. These experiments use other decoders and do not validate FLEX latency or perception.

An ongoing depth study trains independent DCVC-UF intra codecs with two, four and six decoder trunk blocks under the pinned official training recipe. Their encoders and entropy models are also trained, so their latents are not interchangeable exits of the shared-latent model studied here. Released D12 is a deployment reference; a D12 model trained under the same protocol is needed to isolate depth from training provenance. No incomplete training snapshot is included as a final result in this manuscript.

6. Conclusion

Spatial early exit provides a concrete mechanism for reducing decoder arithmetic, but its scientific value is the gain beyond cheaper blind allocations. On the recorded shared-latent DCVC-UF experiment, that gain is largest in a restricted quality-budget range and contracts as the ladder saturates. Delivered-quality checks and the cost of acquiring decision information are equally central to the claim. Together, these observations define a testable route from a flexible decoder to a practical adaptive codec.

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